

EMILY Y. XU

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EDUCATION

California Institute of Technology (Caltech)

Expected Graduation: 06/2027

B.S., Electrical Engineering - GPA: 4.2/4.0

Pasadena, CA

Coursework: Large Language Models for Reasoning, Machine Learning/Data Mining, Deep Learning, Data Structures/Algorithms, Software Design, Digital Logic and Embedded Systems, Deterministic Analysis of Systems and Circuits, Mathematics and Physics of Electrical Engineering, Electronic Systems Lab, Probability and Statistics, Physics E/M, Ethical AI

EXPERIENCE

Machine Learning/Software Engineering Intern

June 2025 – Present

hireEZ

Remote — Mountain View, CA

- Designing and implementing a **Full-stack** job application portal using **React** and **FastAPI**, enabling users to upload resumes, and receive ranked job recommendations based on skill alignment.
- Designing an LLM-powered scoring pipeline leveraging **text embeddings** and **cosine similarity** to match resume content with job descriptions, integrating gpt-4o and interactive UI for resume-job fit exploration.

Summer Undergraduate Research Fellow (Machine Learning)

Feb 2025 – Present

Autonomous Robotics and Control Laboratory, Advised by Prof. Soon-jo Chung and Thomas Berrueta

Pasadena, CA

- Built a physics-based vehicle simulation from throttle input to wheel dynamics, integrating drivetrain, engine, and transmission models with PID-based lateral and longitudinal control.
- Developed and injected realistic disturbance models; trained neural networks to learn residual physics, enabling meta-learning-based adaptive control under shifting conditions.
- Ported Python simulation to **C++** for integration with real-time racecar software, supporting embedded deployment and closed-loop compatibility with existing simulators.

Summer Undergraduate Research Fellow (Physics/Software)

Jun 2024 – Aug 2024

Caltech Observational Cosmology, Advised by Dr. Sunil Golwala and Brandon Sandoval

Pasadena, CA

- Designed aluminum quasi-particle trapping capacitor fins for Kinetic Inductance Detector (KID) devices. Optimized design using circuit theory and quality factor research to improve phonon collection efficiency through current modeling in Sonnet RF simulator.
- Developed algorithms in Python utilizing linear regression and least squares optimization to automate quality factor calculations from Sonnet. Presented to Fermilab and NASA JPL researchers.

PROJECTS

Infinite Vertical Platformer Video Game 🎮 | C, SDL, POSIX API

- Developed a full stack single and multiplayer video game inspired by Doodle Jump from scratch in C, using SDL for rendering.
- Implemented custom physics engine, sprites, keyboard controls, an intelligent enemy AI, UI, and unit testing.
- Utilized POSIX API for system-level programming, optimizing memory management and algorithm efficiency.

LLM Stock Market Analysis Tool 📊 | Python, Beautiful Soup, PostgreSQL, Docker, Flask, Plotly

- Built a Flask web app for real-time stock visualization using Beautiful Soup, Plotly, and PostgreSQL, containerized with Docker.
- Integrated a fine-tuned LLM via Gemini API for market analysis, applying prompt engineering and caching to boost efficiency by 60%.
- Enabled interactive, user-driven insights from scraped financial data with streamlined AI-powered backend.

LEADERSHIP

Mission Lead, Caltech Air and Outer Space

Oct 2023 – Nov 2024

National Aeronautics and Space Association (NASA), Caltech Air and Outer Space (CAOS)

Pasadena, CA

- Directed a team of 10 Caltech students in crafting the mission concept and background research review for a 26-page proposal.
- Secured \$150,000 in funding to advance research on the proposed inflation technology, performed fluids simulations in COMSOL.

Canadian Team Captain

Feb 2022 – Aug 2022

International Young Physicists' Tournament

Timisoara, Romania

- Led a team of 5 in a thesis-style competition solving 17 open-ended physics problems with experimental and theoretical analysis.
- Built a 3-axis Arduino system with MATLAB control and Excel logging; ran EM simulations in COMSOL and Python.

TECHNICAL SKILLS

Programming Languages: Python, Java, C/C++, SQL, ABEL (HDL), $\text{\LaTeX}$

Frameworks & Libraries: TensorFlow, Scikit-learn, PyTorch, Flask, FastAPI, React, Dash, Plotly, Matplotlib, OpenCV, Keras

Tools: MATLAB, PostgreSQL, Docker, Arduino, SolidWorks, COMSOL, Sonnet, DipTrace, Git, VS Code, LTSpice, Foxglove

Skills: Circuit Design, Machine Learning, Deep Learning, Data Analysis, Data Mining, Scientific Writing, Research Methodologies, OOP

AWARDS

International Young Physicists' Tournament: Bronze 2022

Canadian Young Physicists' Tournament: 2x Gold 2022, 2023

Carnegie Mellon University Women's Mathematics Competition: Overall Third Place 2022